

UNIVERSITY INSTITUTE of COMPUTING
Asia's Fastest Growing University

NAAC GRADE A+ ACCREDITED UNIVERSITY

Student Management System

ShivShankar.Kumar

UID :26MCA20195

Abstract

The Student Management System is a C programming project developed to manage student records efficiently. It helps users store, view, search, update, and delete student information. The system reduces manual paperwork and makes record management easier and faster.

Introduction

The Student Management System is a computer-based application used to maintain student information. Managing records manually can be time-consuming and may cause errors. This project uses C programming to simplify student record management and demonstrates concepts such as structures, functions, loops, and conditional statements.

Objectives

To manage student records digitally

To add and display student information

To search for student records easily

To update and delete records when required

To reduce manual work and save time

To apply C programming concepts in a practical project

Software and Hardware Requirements

Software Requirements

Programming Language: C

Compiler: GCC or Turbo C

Code Editor: Visual Studio Code or Code::Blocks

Operating System: Windows or Linux

Hardware Requirements

Computer or laptop

Processor: Intel Core i3 or equivalent

RAM: Minimum 4 GB

Storage: Minimum 500 MB free space

Keyboard and monitor

Implementation

The Student Management System is implemented using the C programming language. Structures are used to store student details, while functions perform different operations. Loops and conditional statements control the program flow. A menu-driven interface allows users to select the required operation. File handling can be used to save student records permanently.

Key Functionalities

Add Student: Enters new student details into the system

Display Students: Shows available student records

Search Student: Finds a student using a roll number

Update Student: Modifies existing student information

Delete Student: Removes a student record

Exit: Closes the program

System Analysis and Differentiation

The Student Management System provides a computerized alternative to maintaining student records manually.

Manual System

Uses paper registers

Searching takes more time

Updating requires manual changes

Requires physical storage

Student Management System

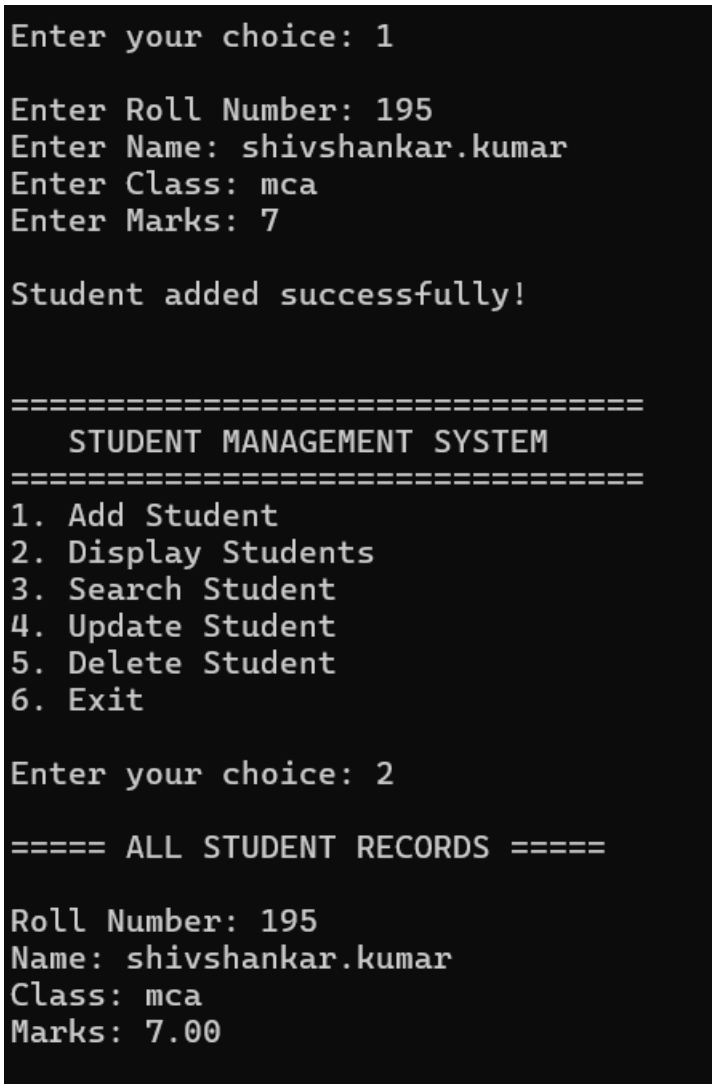
Uses a computer program

Searching can be faster

Records can be updated easily

Stores information digitally

Output and Screenshots



Conclusion

The Student Management System developed using C programming provides a simple way to manage student records. It helps organize information, reduces manual work, and improves practical understanding of C programming concepts.

Future Advancements

Add a secure login system

Include student attendance management

Add examination and marks management

Generate student reports automatically

Improve data validation and security

Implement file handling for permanent storage

Add backup and recovery facilities